

Section 1.7 HW. Write-up

3. Prove that the product of an odd integer and an even integer is odd. (If x is odd and y is even, then xy is even)

Proof

Let x be an odd integer and y be an even integer. [show xy is even]

$x = 2k+1$ and $y = 2l$ for some integers k and l .

Then $xy = (2k+1)(2l) = 4kl + 2l = 2(2kl + l)$ is even since $2kl + l$ is an integer whenever k and l are integers



9. If n is odd, then $n^2 = 8i+1$ for some integer i .

Proof

Let n be odd. So $n = 2k+1$ for some integer k .

Then $n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 4k(k+1) + 1$.

Consider 2 cases. By the parity principle k is either even or odd.

Case 1: If k is even we have $k = 2l$ for some integer l so

$n^2 = 4k(k+1) + 1 = 4(2l)(k+1) + 1 = 8(l \cdot (k+1)) + 1$ has form $8i+1$ where i is the integer $l(k+1)$

Case 2 If k is odd then $k+1$ is even. We can replace $k+1$ by $2m$ for some integer m .

$n^2 = 4k(k+1) + 1 = 4k(2m) + 1 = 8(km) + 1$ has form $8i+1$ where i is the integer km .



12. The sum of two rational numbers is rational.

(If x and y are rational, then $x+y$ is rational.)

Proof:

Let x and y be rational.

This means $x = \frac{a}{b}$ and $y = \frac{c}{d}$ where a, b, c, d are integers and $b \neq 0, d \neq 0$.

$x+y = \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ is rational since $ad+bc$ and bd are integers and $bd \neq 0$ by the zero product property.

□

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We disprove this by a counterexample.

$x = 0$ is rational $y = \sqrt{2}$ is irrational but

$x+y = 0+\sqrt{2} = \sqrt{2}$ is not rational.

□

Note: If x is rational and y is irrational, then $x+y$ must be irrational. See example 1.7.2 on page 72